Network Position and Academic Performance

Adding Network Centrality to Smirnov & Thurner (2017)

ECC3841 Network Economics — Research Brief

Intro

A student's grades depend on more than effort and ability. They also depend on the social world around that student. Many studies show that friendship networks affect GPA, but how they affect it is still an open question.

One well-known study, by Smirnov and Thurner (2017), used a real, changing "who-likes-who" network of Russian high-school and university students. They found that friendship's effect on GPA is mostly about **selection**: students change friends to match their own GPA, instead of being pulled up or down by their friends' grades. Their evidence is strong. But their method only ever asks one question: who are this student's friends, and what is the average GPA of those friends? It never asks where a student sits inside the wider network. Is this student a hub that many people like on their own? Or can this student only be reached through long, indirect chains that pass through many different social groups? That second question, whether **network position**, not just who your friends are, affects grades on its own, is what this project looks at. We use the same public data that Smirnov and Thurner released.

Research Question

Does your position in a friendship network affect your grades?

Main Idea

The "traditional" (Smirnov & Thurner) idea:

Friend selection $\rightarrow$ friend-group GPA similarity $\rightarrow$ observed homophily in GPA

Their method asks one thing: who is this student connected to, and what is the average GPA of those people? Two students with the same direct friends would get the same score under this method. It would not matter how different those friends' own social worlds are.

What we aim to analyse:

Network position (direct popularity vs. indirect reach) $\rightarrow$ academic performance, net of friend-group composition

Say two students have the same size friend group, with the same average GPA. Smirnov and Thurner's method would treat these two students as the same.

- **Student 1** is liked by many classmates directly. This popularity sits inside one close, tight circle of friends.
- **Student 2** is liked by fewer people directly. But through friends of friends, Student 2 connects to many different, loosely-linked groups across the school. Their reach is wide, even though their direct popularity is nothing special.

Should we expect Student 1 and Student 2 to end up with the same GPA, just because their friend group's average GPA looks the same? Our answer, below, is **no**. And the direction of the difference is not what you would guess if you assumed "more connections is always better."

Predictions

1. We expect to repeat Smirnov and Thurner's two results: (a) once we account for a student's own past GPA, a friend's past GPA will not predict that student's future GPA, which would rule out direct peer pressure on grades; (b) new friendships will have a smaller GPA gap than friendships that end, a sign of selection.
2. Katz-Bonacich centrality, a network-position measure the original paper never used, will still help predict GPA after we control for the friend-composition effect Smirnov and Thurner found.
3. The selection effect (prediction 1b) will be stronger in the university sample than in the high-school sample. University students have more freedom to change their social circle: new courses, clubs, and dorms all give them fresh chances to make new friends.
4. The network-position effect from prediction 2 will be stronger in the high-school sample than in the university sample. High-school friend groups are more stable and build up over more time.

Model Structure

Nodes: individual students (655 high-school students; 1,200–1,549 students per university group — freshmen, sophomores, juniors, seniors).

Edges: directed "like" ties, taken from a social media site. An edge from i to j means i gave j at least one like within a 3-month period. We observe 2 to 14 network snapshots per group (38 in total). These networks are **not symmetric**: only 24–27% of ties go both ways, so a "friend" in this data really means "someone I chose to like," not a confirmed two-way friendship.

Node attributes:

1. GPA (the outcome; it changes over time for the high-school group, but is a single measurement for the four university groups)
2. Network position: in-degree (direct popularity — how many people liked this student), out-degree, Katz-Bonacich centrality (computed at several decay levels), betweenness, eigenvector centrality
3. Group / level: high school vs. university (and the specific class-year)

Related Studies

1. AddHealth-based peer effects (IZA DP 3859). *Nodes:* individual students. *Edges:* real friendship nominations from AddHealth — each student named their best friends from a school roster. *Node attribute:* school performance / GPA, the main outcome.

2. Peer depression and long-term mental health (Giulietti, Vlassopoulos & Zenou, 2020, IZA DP 13680, "Peers, Gender, and Long-Term Depression"). Studies whether exposure to peers' depression in adolescence predicts a person's own depression in adulthood, using AddHealth data. The effect holds for females but not males: a one-standard-deviation rise in the share of depressed same-gender schoolmates raises the odds of adult depression by 2.6 percentage points for women. Family socioeconomic background shows up as a *moderator*, not the main explanatory variable: girls from lower-SES families are more exposed to this peer effect, while a stronger family background acts as a buffer. *Nodes:* individual students. *Edges:* shared membership in the same school–cohort–gender reference group (a complete network within each group), not individual friendship ties. *Node attribute:* adult depression status. One of the authors, Yves Zenou, teaches this course.

3. Smirnov, I., & Thurner, S. (2017). "Formation of homophily in academic performance: Students change their friends rather than performance." ***PLOS ONE*, 12(8): e0183473.** *Nodes:* individual high-school and university students (a Russian sample of about 6,200 students). *Edges:* directed "like" ties from a social media site, observed every 3 months over up to 3.5 years. *Node attribute:* GPA, the main outcome. *Method:* (i) a regression showing that a friend's past GPA does not predict a student's future GPA, once we control for the student's own past GPA, which rules out direct peer influence; (ii) a comparison of GPA gaps for new ties versus dropped ties, showing that selection, not influence, explains why friend groups end up with similar GPA; (iii) a simple agent-based model that reproduces this pattern over time. **This is the dataset and the paper our project builds on and replicates.** Its main limit, and the gap our project fills, is this: the network is only ever used to compute one simple number, the average GPA of a student's direct friends. It never looks at the wider shape of the network, so it cannot tell "popular" apart from "well-reached."

The Full Chain of Reasoning

This section walks through the actual path we took, dead ends included, because the corrections along the way are part of what makes the final conclusion solid.

Step 1 — Replicate the original paper on its own data

We used the same data (Harvard Dataverse, doi:10.7910/DVN/SZA9YW) and rebuilt Smirnov and Thurner's two main checks in Python. Both checks came out the same way as in the original paper. First, we regressed GPA at time t on GPA at time $t-1$ and average friend GPA at time $t-1$, using the high-school group only, since it is the one group with a real GPA time series. Once we controlled for a student's own past GPA (coefficient 0.955, $p < 0.001$), the friend-GPA term had no effect (coefficient -0.0055, $p = 0.557$). So a friend's past grades do not predict a student's future grades. Second, we compared GPA gaps for new ties and dropped ties. In the university sample, new ties had a smaller GPA gap than dropped ties (difference -0.021, $p < 0.001$). In high school, the same pattern showed up but was not statistically significant (-0.010, $p = 0.183$). This supports Prediction 3: university students have more freedom to change friends, and that shows up as a stronger, cleaner selection effect.

Step 2 — Add Katz-Bonacich centrality; get an encouraging but naive result

Next we added Katz-Bonacich centrality (computed per network snapshot, with $\alpha = 0.85 \times 1/\lambda_{max}$) to a regression of GPA on average friend GPA and group fixed effects. The centrality coefficient was positive and significant (standardized $\beta = 0.093$, $p < 0.001$). At first glance, this looked like clean support for Prediction 2.

Step 3 — Stress-test it: does this survive a check against raw degree?

Katz-Bonacich centrality correlates 0.85–0.89 with plain in-degree in these networks. It might just be a complicated way of asking "how many people like you?" So we ran the same regression again, this time with raw in-degree and out-degree added alongside Katz-Bonacich. The Katz coefficient flipped sign and lost most of its significance, dropping from +0.093 ($p < 0.001$) to a small, unstable number. In-degree stayed strongly significant, and R^2 barely changed. **The result from Step 2 does not survive this test.** It was mostly popularity wearing a disguise. We report this as a real negative finding, not something to hide.

Step 4 — Split Katz-Bonacich apart by its decay parameter

Katz-Bonacich centrality has a decay parameter, α, that we can adjust. When α is close to 0, the measure is almost the same as raw in-degree. When α gets close to its upper limit ($1/\lambda_{max}$), the measure increasingly reflects long, indirect paths through the network. We tested α at six levels, from 5% to 95% of that upper limit. As α rose, the correlation with in-degree fell from 0.9999 down to 0.753. More importantly, the Katz coefficient, tested against raw degree each time, moved through three stages. At low α, it was positive, but this was just an artifact of being almost the same as degree ($\beta = +0.082$, $p = 0.012$). In the middle range, it dropped out of significance. At high α, it turned into an independent, negative effect ($\beta = -0.048$, $p = 0.031$ at $\alpha = 0.80$; $\beta = -0.036$, $p = 0.011$ at $\alpha = 0.95$). Betweenness centrality, a different kind of "bridge" measure that is less tied to degree ($r = 0.70$, versus 0.87 for Katz), showed this same negative, independent effect when we tested all four centrality measures together ($\beta = -0.059$, $p < 0.001$). Eigenvector centrality, like Katz at low α, was fully absorbed by degree.

α (frac. of max)	0.05	0.20	0.40	0.60	0.80	0.95
Corr. with in-degree	0.9999	0.9976	0.9884	0.9657	0.9053	0.7530
Katz coef. (vs. degree)	+0.082	+0.022	−0.034	−0.051	−0.048	−0.036
p-value	0.012	0.598	0.407	0.115	**0.031**	**0.011**

Table 1: As α rises, Katz-Bonacich moves away from raw popularity, and an independent negative effect appears.

Step 5 — Lock down one pre-set headline test

Step 4 tested six levels of α. If we said "significant at 4 of 6 levels," that would overstate our evidence, because nearby α values are closely related to each other. This is closer to one smooth trend crossing a line than four separate, independent tests. To avoid overstating our result, we treat one test as the main, confirmed result: the $\alpha = 0.85$ setting we chose before running the sweep, back when we first built the dataset. We test this once, using the full sample, and control for raw in-degree, out-degree, average friend GPA (Smirnov and Thurner's own variable), and group fixed effects, all at the same time:

$$GPA_{it} = \beta_0 + \beta_1 \,\mathrm{Katz}^z_{it} + \beta_2 \,\mathrm{InDegree}^z_{it} + \beta_3 \,\mathrm{OutDegree}^z_{it} + \beta_4 \,\overline{GPA}^{friends}_{it} + \gamma_{\mathrm{group}} + \varepsilon_{it}$$

Katz-Bonacich coefficient $\hat{\beta}_1 = -0.0415$ (SE $= 0.019$, $p = 0.033$, clustered by student, $n = 36{,}696$).

Average friend GPA is already in this regression. This is the same variable that carries Smirnov and Thurner's selection effect. So if Katz still has a significant effect on top of that, it means network position carries information their method cannot see. It is not just their finding wearing a new name.

Step 6 — Test Prediction 4 properly, across the whole α range

At our fixed $\alpha = 0.85$, only two of the five groups showed a significant effect: high school ($\beta = -0.085$, $p = 0.021$) and juniors ($\beta = -0.100$, $p = 0.008$). The full "university" group, taken as a whole, was not significant at that one α level. Running the same test across the whole α sweep gave a fuller picture:

Group	α=0.05	α=0.20	α=0.40	α=0.60	α=0.80	α=0.95
High school	−0.010	−0.075	**−0.129***	**−0.128***	**−0.099***	**−0.071***
Juniors	+0.029	−0.107	**−0.194***	**−0.194***	**−0.156***	**−0.110***
Freshmen	−0.087	−0.198	−0.118	−0.064	−0.021	+0.012
Sophomores	+0.059	+0.112	+0.120	+0.103	+0.078	+0.046
Seniors	**+0.194***	**+0.190***	+0.109	+0.039	−0.003	−0.021

Table 2: Katz coefficient, tested against degree, for each group across the full α sweep. $^*p < 0.05$.

High school is negative and significant at four of the six α levels. This is the most consistent result in the whole dataset, and it corrects an earlier, too-quick reading that said high school had no effect at all. Pooled university, on the other hand, is only significant at the extreme high end ($\alpha = 0.95$: $\beta = -0.032$, $p = 0.032$). Breaking university down by year shows this is not really one clean "university" story. Juniors match the high-school pattern closely. Sophomores run the opposite way at every single α level; they are never significant, but they never once cross zero either. Freshmen and seniors give mixed results, likely because we do not have enough data for them (freshmen have only 2 network snapshots). **So we reject Prediction 4 as we first stated it.** The pattern does not split cleanly along high-school versus university lines. It depends on the specific group — and Step 7 gives a possible, testable reason why.

Step 7 — A possible reason for the group-to-group differences

Only two of the five groups back the finding on their own. Rather than treat this as a weakness to hide, we asked what separates those two groups from the other three — and found a pattern worth reporting, not just a correlation to wave at.

We ranked the five groups by average in-degree, i.e. how *sparse* each network is on average. Then we compared this to each group's Katz coefficient at $\alpha = 0.95$. A clear pattern showed up. The two groups with the strongest negative effect, high school (average in-degree 3.84) and juniors (3.95), are the two **sparsest** networks in our data. The two groups running the opposite way, sophomores (5.06) and freshmen (5.61), are among the **densest**. Across all five groups, this correlation is $r = 0.86$. With only five points, this is not enough for a formal statistical test, but it points to a possible reason, stated in plain terms:

> In a social circle that is sparse overall — where the average student does not have many direct connections — reaching far beyond your direct friends, through several loosely-connected steps, is itself costly: it can only be achieved by genuinely spreading your attention thin across separate, disconnected pockets of people. In a dense circle, the same "wide reach" may just mean being part of one big, already well-connected group, at little extra cost. That would explain why the penalty for wide indirect reach shows up in the two sparsest networks (school, juniors) and not in the denser ones (sophomores, freshmen).

To be clear about what this is and is not: this is an explanation we *infer* from the pattern across five group-level data points, not a mechanism we directly tested (we do not, for instance, measure students' actual time spent maintaining ties). It is a plausible, testable hypothesis for future work, not a proven causal channel — and we report it as exactly that.

Step 8 — Compare the effect size honestly

We recomputed Smirnov and Thurner's own main statistic: the Homophily Index, the Pearson correlation between a student's GPA and their friends' average GPA. Using their exact published formula on the same data, we get a pooled average of $r \approx 0.315$ (by group: school 0.304, freshmen 0.250, sophomores 0.298, juniors 0.303, seniors 0.344), consistent with a medium-to-strong, well-established effect. The regression-scale analogue in our own headline model, the coefficient on $\overline{GPA}^{friends}$, is **0.476** GPA points per GPA point of friend average, a large, dominant effect. Against that backdrop, our headline Katz effect of $-\mathbf{0.0415}$ standard deviations is genuinely small, on the order of one-tenth the size of the selection channel the original paper already established.

Conclusion

Confirmed: the original paper's main selection-versus-influence logic holds up on this data (Prediction 1). University's greater social flexibility does produce a stronger, measurable selection effect than high school's (Prediction 3).

Revised: network position does carry information beyond friend-group makeup (Prediction 2), but only once we correctly separate "network position" from raw popularity using the decay parameter. A simple, naive centrality regression does not hold up to scrutiny. **Rejected:** the effect does not split cleanly along high-school versus university lines (Prediction 4). Only two of five groups (high school, juniors) back it on their own — the two **sparsest** networks in the data ($r = 0.86$ between a group's average in-degree and its Katz coefficient). We report this as a boundary condition on the finding, not as noise: the effect looks like something that shows up when social circles are sparse enough that reaching widely has a real cost, and disappears when circles are dense enough that wide reach is nearly free.

Our main finding, stated plainly: being directly liked by many people (popularity), and being reachable through many indirect, loosely-connected paths (wide network reach), are two separate things. We can tell them apart in the data, and they point in opposite directions for GPA. Popularity helps. Wide reach mildly hurts, once we hold popularity fixed — and only in the sparser social settings we studied. Smirnov and Thurner's method could never have found this, because it only ever measures a student's direct friends' average GPA. The effect is modest, roughly a tenth the size of the friend-selection effect the original paper found, and it does not show up in every group we tested. We report both of these limits directly, instead of hiding them.

Why this still matters. Not because of the size of one number, but for three reasons. First, we show a working way to separate a network's structural position effect from its friend-group composition effect. This is a hard problem in economics: telling apart what your position in a network does to you, from who you already chose to be friends with. We make progress on it using a tool the original paper's method could not access. Second, we built a method other people can reuse: split a centrality measure by its own decay parameter to pull apart local popularity from extended reach. This could be applied to other networks too. Third, our finding pushes back on a common assumption that people rarely question, that a "bigger network" or "more connections" is always good. We give a specific, testable counter-example. Reach without depth is not obviously a good thing — and in the sparsest networks in our data, it is a measurable cost.

What this study cannot claim. This is one dataset: one country, one time period, one platform's definition of a "like." A robust, well-tested pattern in one real dataset is not the same as a universal law, and we do not claim one. Whether this pattern holds outside Russian students on VK is a question for future, independent data — not something this project can answer on its own.

Data: Smirnov & Thurner (2017) replication data, Harvard Dataverse, doi:10.7910/DVN/SZA9YW. Full analysis pipeline: `scripts/friendship_gpa/01--04`. All regression outputs referenced above are saved under `output/friendship_gpa/`.